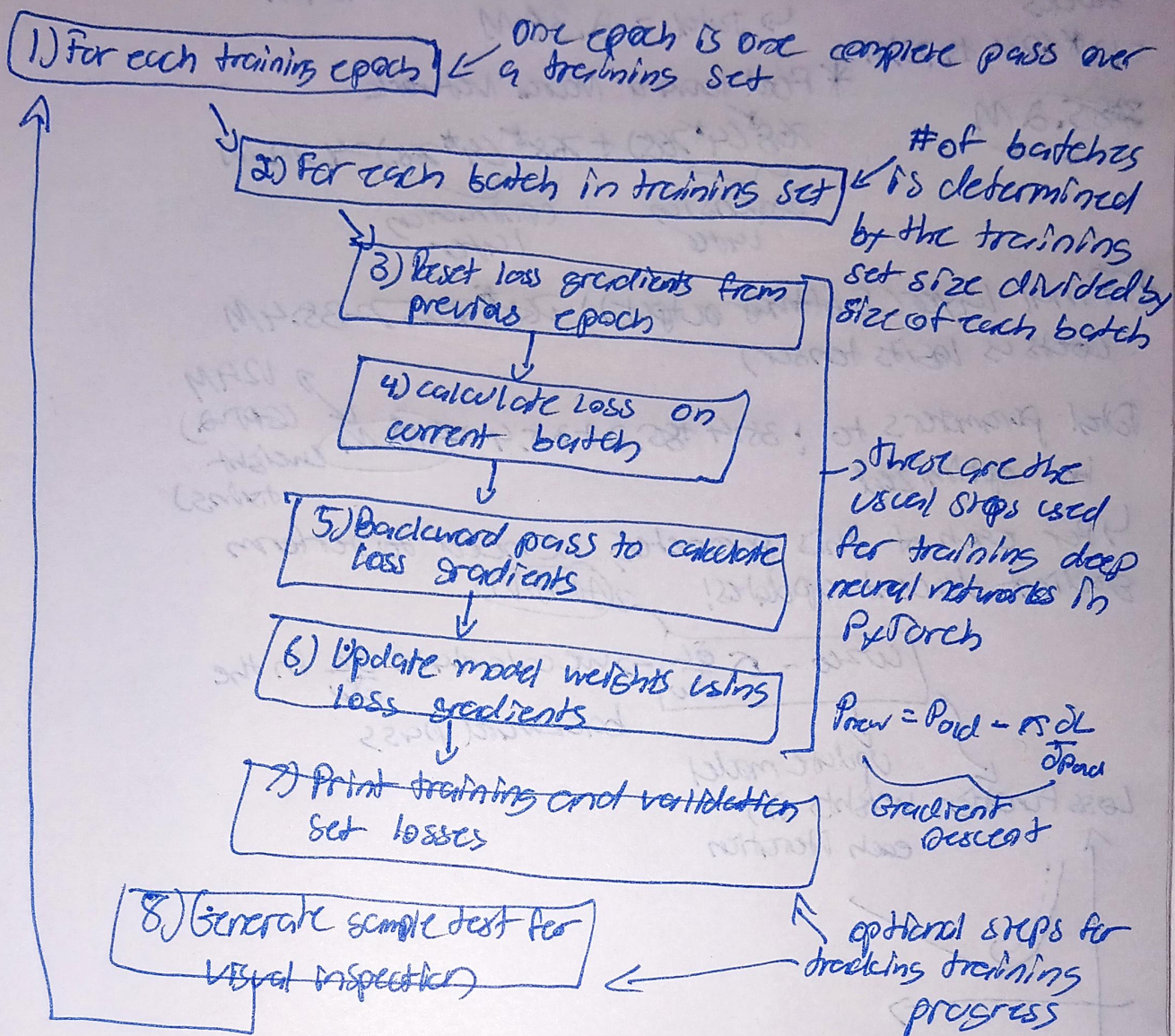
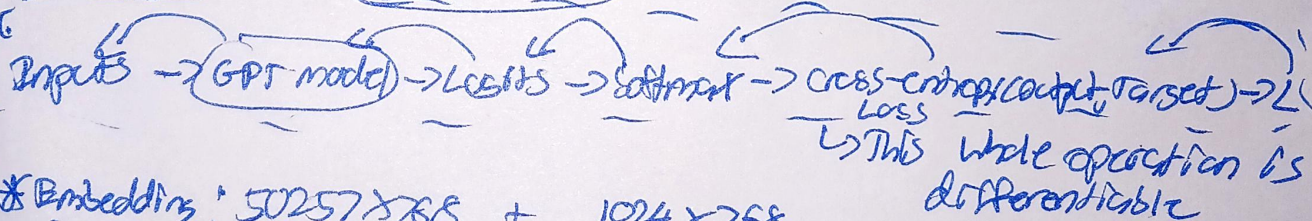


How to train the LLMs → LLM training function

LLM pretraining loop schematic → outputs = target



Main Step: Find loss gradients using loss.backward()



* Embedding parameters: 50257×768 (Token) + 1024×768 (Positional) = 38.4M

→ NEXT PAGE

* Transformer Block! * Multi-head attention!

Total for 12 transformer blocks

$\hookrightarrow$ Ques key value = $3 \times 768 \times 768 \approx 1.72M$
 $\hookrightarrow$ output head = $768 \times 768 = 0.59M$
 $\hookrightarrow$ total = $2.36M$

$\approx 12 \times 6236 \approx 4.72M$

$\approx 85.2M$

* Feed forward Neural Network

$768 \times (4 \times 768) + 768 \times (4 \times 768) = 4.72M$
 Expansion later contraction later

* Final layer (Softmax output): $768 \times 50257 = 38.4M$
 (6236 is logits tensor)

Total parameters to be optimized: $38.4 + 85.2 + 38.4 \approx 162M$

Per each of this parameters we need to perform gradient descent updates!

ADAM

$w_{new} = w - \alpha \frac{\partial L}{\partial w}$

$\rightarrow$ we calculate $\frac{\partial L}{\partial w}$ in the backward pass

Update model weights in each iteration

